9. Challenges and Learning

- **Challenge encountered:** One important challenge was making sure that the analysis used consistent rules and did not produce misleading results when the input data was invalid. For example, a student must fail if even one subject mark is below 40, regardless of the overall average. Missing values, duplicate student IDs, non-numeric entries, and marks or attendance outside the 0-100 range also needed to be handled before calculations could be trusted.
- **How it was addressed:** I organised the project into reusable functions so that validation, calculations, and summaries could be handled separately. The `validate_student_data()` function checks the required columns, missing and blank values, duplicate IDs, numeric fields, and allowed ranges. The `determine_pass_fail()` and `assign_grade()` functions apply the pass/fail and grading rules consistently. I also used Python's `unittest` framework to check pass/fail conditions, grade boundaries, student calculations, class and subject summaries, duplicate IDs, and out-of-range marks. Finally, I ran the main program and test suite against the supplied dataset to verify that the analysis completed successfully.
- **What I learned:** I learned how to load CSV data into a Pandas DataFrame, inspect and validate records, and use functions, loops, and conditions to organise a Python data-analysis workflow. I developed a clearer understanding of using NumPy for totals, averages, and numerical checks, and Pandas for descriptive statistics, filtering, sorting, and summarising student and subject performance. I also learned why input validation, clearly defined business rules, and testing edge cases are important for reliable results. The project helped me practise turning raw tabular data into understandable summaries using the Python concepts required for the assignment, without needing machine learning or prediction.